

Assignment 8



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92% (23/25)

Q1(a,b)

6 / 6

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a.

$$\sum_{n=0}^{\infty} \frac{3^n}{e^{n-1}} = \sum_{n=0}^{\infty} \frac{3^n}{e^n} e = e \sum_{n=0}^{\infty} \left(\frac{3}{e}\right)^n$$

Since $\frac{3}{e} > 1$, $\lim_{n \rightarrow \infty} \left(\frac{3}{e}\right)^n = \infty \neq 0$. Therefore, the series is **divergent** by the **divergence test**.



b.

$$\sum_{n=1}^{\infty} \frac{1+2^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n + \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$$

Both are **convergent** since they are **geometric series**. Sum can be evaluated using $\frac{a}{1-r}$

$$\begin{aligned} \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n &= \frac{1/3}{1-1/3} = \frac{1}{2} \\ \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n &= \frac{2/3}{1-2/3} = 2 \end{aligned}$$

Total sum:

$$\sum_{n=1}^{\infty} \frac{1+2^n}{3^n} = \frac{1}{2} + 2 = \boxed{\frac{5}{2}}$$

Q1(c)

3 / 4

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c.

$$\frac{1}{n^2 + 7n + 12} = \frac{1}{(n+3)(n+4)} = \frac{1}{n+3} - \frac{1}{n+4}$$

The series is **convergent** since terms cancel out:

$$\begin{aligned} \sum_{n=1}^{\infty} \left(\frac{1}{n+3} - \frac{1}{n+4} \right) \\ &= \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) + \left(\frac{1}{6} - \frac{1}{7} \right) + \dots \\ &= \frac{1}{4} + \left(\frac{1}{5} - \frac{1}{5} \right) + \left(\frac{1}{6} - \frac{1}{6} \right) + \dots \\ &= \boxed{\frac{1}{4}} \end{aligned}$$

Need more details on
steps

-1

Q1(d)

3 / 3

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d.

$$\ln \left(\frac{n}{n+1} \right) = \ln n - \ln(n+1)$$

Terms cancel out:

$$\begin{aligned} S_N &= \sum_{n=1}^N (\ln n - \ln(n+1)) \\ &= (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + (\ln 3 - \ln 4) + \dots \\ &= \ln 1 + (-\ln 2 + \ln 2) + (-\ln 3 + \ln 3) + \dots - \ln(N+1) \\ &= \ln 1 - \ln(N+1) = -\ln(N+1) \end{aligned}$$

Taking the limit:

$$\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} -\ln(N+1) = -\infty$$

The last term diverges, therefore the series is **divergent**.

Q2

4 / 4

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After the n^{th} bounce the ball rises to a height of $8\left(\frac{2}{3}\right)^n$, therefore:

$$D = 8 + 2 \sum_{n=1}^{\infty} 8\left(\frac{2}{3}\right)^n$$

The series is **convergent** by the **geometry series test**, and has sum:

$$\sum_{n=1}^{\infty} 8\left(\frac{2}{3}\right)^n = \frac{8 \cdot \frac{2}{3}}{1 - \frac{2}{3}} = \frac{16/3}{1/3} = 16$$

Therefore the total distance traveled is:

$$D = 8 + 2(16) = \boxed{40 \text{ m}}$$

Q3(a)

3 / 3

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a.

Use **integral test** with $f(x) = \frac{1}{x \ln x}$. On $[2, \infty)$ this function is **positive, continuous, and decreasing**. Using the substitution $u = \ln x$, $du = \frac{1}{x} dx$:

$$\int_2^{\infty} \frac{1}{x \ln x} dx = \int_{\ln 2}^{\infty} \frac{1}{u} du = \ln u \Big|_{u=\ln 2}^{u=\infty} = \infty$$

The integral diverges, therefore the series is **divergent**.

See Learn.

b.

Use *integral test* with $f(x) = \frac{1}{x(\ln x)^p}$, which is *positive, continuous, and eventually decreasing* for any p . Using the same substitution $u = \ln x$:

$$\begin{aligned} \int_2^\infty \frac{1}{x(\ln x)^p} dx &= \int_{\ln 2}^\infty \frac{1}{u^p} du \\ &= \int_{\ln 2}^\infty u^{-p} du \\ &= \frac{u^{1-p}}{1-p} \Big|_{u=\ln 2}^{u=\infty} \\ &= \begin{cases} \infty \\ (\ln 2)^{1-p}/(p-1) \end{cases} \end{aligned}$$

Therefore, the series *converges* for $p > 1$ and *diverges* for $p \leq 1$.

This proof
doesn't work
for $p = 1$

-1